
Specialist Physical AI Need Not Be Opaque

A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control, failure analysis, and weight-derived causal structure

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Abstract

1 Tensor decomposition turns a large table of learned interactions into simpler low-
2 rank factors that can be ranked and inspected. Conventional robot policies con-
3 tain operations that block this direct weight-level view. We introduce χ -VLA, a
4 20.1M-parameter vision-language-action policy built only from bilinear, linear, and
5 rational learned operations. Its weights can therefore be reorganized into explicit
6 interaction tensors rather than treated only as an opaque checkpoint. A mechanical
7 audit and numerical reconstructions verify that the deployed policy preserves this
8 form. Trained from scratch on LIBERO-Object, convolutional χ -VLA reaches
9 $93.7\% \pm 0.8\%$ over three seeds under the published protocol. We reconstruct
10 every attention module in three checkpoints with worst relative error 2.56×10^{-7} ,
11 and exact weight-derived subspaces beat equal-rank random controls in all 36
12 primary block-head tests. Retaining 64 of 384 causally ranked visual dimensions
13 preserves 18/20 successes, while a random subspace preserves none. A 300-pair
14 instruction intervention reliably changes motion but rarely overcomes the scene
15 prior completely, and the first direct coefficient edit fails its selectivity gate. A
16 same-skeleton conventional twin differs by only 0.006% in parameters. Complete
17 same-hardware replications find a 4.4-point rational deficit on A30 and 5.8 points
18 on A6000. Compilation reduces the same-A30 latency ratio from $3.49\text{--}5.33\times$
19 eager to $1.73\text{--}1.74\times$ compiled. The result is a capable and structurally inspectable
20 specialist, not a zero-shot generalist, a compute-matched system, or a solved editing
21 interface.

22 1 Introduction

23 Modern robot policies can achieve high success while giving little indication of why an action was
24 chosen. Their checkpoints contain millions of weights, but those weights do not directly show which
25 image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated
26 activations, yet a correlation is not automatically a mechanism or a safe editing target.

27 A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of
28 low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses
29 additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction
30 table defined by its weights. Its decomposed factors then provide candidate control mechanisms that
31 can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully
32 tensor-decomposable policy.

33 The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization
 34 break the algebraic form, while continuous actions require precise closed-loop outputs. The potential
 35 payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows
 36 language, keys on a visual shortcut, or uses a fragile control feature.

37 We study the narrower but testable question:

38 *Can a specialist VLA retain competitive closed-loop capability when every deployed*
 39 *learned operation is polynomial or rational, and does the resulting structure*
 40 *provide a useful weight-based analysis surface?*

41 The stronger thesis asks for three results on the same policies: small capability cost, stable weight-
 42 derived causal structure, and selective edits without retraining. We establish the first two only partially
 43 and report the failed first surgery screen rather than treating algebraic representability as proof of
 44 interpretability.

45 Our contributions are:

- 46 1. We construct a complete VLA from tensor-compatible primitives. A rational normalization
 47 supplies input-specific scaling without leaving the rational function class.
- 48 2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with
 49 published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm$
 50 0.8% over three seeds, and a three-checkpoint ensemble reaches 96.2%. A same-skeleton
 51 conventional twin differs by only 0.006% in parameters. Two complete same-hardware
 52 replications place the seed-0 capability cost between 4.4 and 5.8 points.
- 53 3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational
 54 and bilinear branches from their weights.
- 55 4. We extend exact weight-derived decomposition through individual attention layers with a
 56 reduced-Gram calculation that avoids materializing the full high-order tensor.
- 57 5. We show a causal closed-loop consequence. A selected 64-dimensional visual subspace
 58 preserves almost all tested behavior, while a random equal-width subspace fails. A separate
 59 three-checkpoint instruction intervention distinguishes reliable trajectory response from
 60 incomplete task-level rerouting.

61 This paper does not claim current state of the art. It also does not claim that comparable mechanisms
 62 are impossible to obtain from conventional policies. The empirical question is whether the algebraic
 63 policy makes them more direct, stable, and selectively editable than matched post-hoc methods.

64 2 A rational tensor robot policy

65 2.1 Tensor-compatible primitives

66 Each primitive replaces a familiar transformer component while keeping explicit coefficients. Lin-
 67 ear maps contribute first-order terms, while multiplying learned projections creates higher-order
 68 interactions that remain determined by the weights.

69 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
 70 layer to represent bias, linear, and quadratic terms in one contraction.

71 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

72 The inner sum is a table indexed by one output and two input coordinates. The three matrices give its
 73 CP factorization, the tensor analogue of a low-rank matrix factorization.

74 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

75 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
76 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
77 interaction coefficients determined by the weights.

78 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
79 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

80 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
81 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
82 rescaling without leaving the rational tensor representation.

83 2.2 Architecture and training

84 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
85 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
86 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
87 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
88 continuous action dimensions. The complete model has 20,137,352 learned parameters.

89 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
90 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
91 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
92 that verify every requested state was loaded.

93 2.3 Mechanical certificate on the trained checkpoint

94 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
95 point at runtime and reconstruct representative operations from saved weights.

96 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
97 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
98 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
99 summarized in Appendix Figure 8. It does not materialize one compact polynomial for the full
100 eight-layer transformer.

101 3 Protocol-aligned closed-loop capability

102 3.1 Evaluation protocol

103 We evaluate all ten LIBERO-Object tasks with:

- 104 • 50 canonical trials per task
- 105 • a 280-step cap
- 106 • three independently trained seeds per architecture
- 107 • 500 rollouts per seed and 1,500 rollouts per architecture

108 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
109 rerun in our codebase, so differences may still include implementation and training choices.

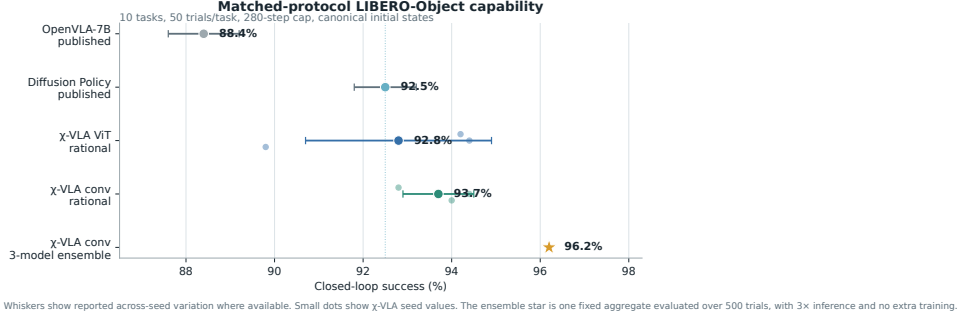


Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational ViT	20.1M	3	92.8% $\pm$ 2.1%
Same-skeleton conventional ViT	20.1M	1	89.8%
χ -VLA, rational conv	20.1M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 20.1M$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The ViT seeds score 89.8%, 94.4%, and 94.2%. The convolutional seeds score 94.0%, 94.4%, and 92.8%. The convolutional model is modestly higher and more consistent in this three-seed comparison.

The same-skeleton conventional twin replaces the rational blocks with softmax attention, SwiGLU, and RMSNorm while holding the data, updates, seed, decoder, and 500 canonical trials fixed. On A30, the conventional twin succeeds in 448/500 trials and the rational checkpoint in 426/500. On A6000, they reach 451/500 and 422/500. Their parameter counts differ by only 0.006%. The rational deficit is 4.4 and 5.8 points, and task 3 accounts for 21 lost successes on each platform. On one A30, eager latency is 38.05 versus 7.14 ms. Compilation reduces it to 1.52 versus 0.88 ms, a $1.74\times$ gap. Rational training is $3.12\times$ slower at batch 128 and uses $1.63\times$ the peak allocated memory. A6000 profiling confirms a $1.73\times$ compiled ratio.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. Seed 0 misses the predeclared three-point noninferiority margin on both GPU classes. A multi-seed claim still requires conventional seeds 1 and 2 and partial-conversion controls. The result isolates a modest capability cost and a remaining systems cost.

131 4 Exact weight-derived structure through attention

132 4.1 Exact layerwise construction

133 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
134 a coefficient table whose entries measure how combinations of query, key, and value coordinates
135 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
136 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
137 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

138 The deployed attention also normalizes each query and key with Equation 3. Each normalization
139 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
140 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
141 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
142 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
143 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
144 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
145 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

146 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
147 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
148 rank important directions without constructing the full table. It matches brute-force materialization to
149 2.13×10^{-14} at toy scale.

150 4.2 Trained policy result

151 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
152 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
153 sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached
154 activations are used only afterward to score how well it preserves the layer’s computation.

155 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
156 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
157 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
158 0, 6, and 7.

159 This extends exact weight-only construction through individual attention layers. It is not an exact
160 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
161 rapid degree and representation growth.

162 5 Causal policy structure and a language shortcut

163 We next test whether the identified structure affects closed-loop behavior and whether high task
164 success reflects robust use of the language input.

165 5.1 Causal visual subspace

166 At the visual bond before the joint backbone, we rank directions separately for translation, rotation,
167 and gripper actions. A subspace is a smaller coordinate system inside the model’s 384-dimensional
168 visual representation. Retaining it means zeroing all other directions, then running the actual
169 policy. We compare selected and random equal-width subspaces. Because the ranking uses held-out
170 activations and downstream sensitivity, it tests a causal bottleneck but is not a weight-only discovery
171 result. With 64 of 384 dimensions retained, the selected subspace reduces action reconstruction error
172 relative to random by $4.8\times$ for translation, $4.6\times$ for rotation, and $41.8\times$ for the gripper. Appendix
173 Figure 4 reports the full comparison. In closed loop, the full representation succeeds in 19/20 trials,
174 the selected subspace in 18/20, and a random subspace in 0/20.

175 This result establishes a causal low-dimensional substrate, but it is still limited to one checkpoint and
176 twenty rollouts. The current ranking at this full-policy bond uses held-out examples and downstream
177 sensitivity. The exact layerwise attention method in Section 4 must still be connected to the same
178 full-suite closed-loop intervention.

5.2 Counterfactual instruction test

High success does not guarantee robust language use. A one-step screen responds to a renamed visible object in only 15.0% of 80 states. We therefore run 100 paired 80-step rollouts for each of three checkpoints. Relative end-effector preference for the renamed object increases by 0.175, 0.168, and 0.189 m, with all trial-bootstrap 95% intervals above zero. Motion shifts toward that object in 84%, 86%, and 88% of pairs, yet only 33%, 28%, and 28% end closer to it. Appendix Figure 3 shows the replication. The seed-0 conventional twin has a 0.239 m shift, 92% directional response, and the same 33% endpoint rate as rational seed 0. Language response is therefore not unique to tensor decomposability. This is a reach-phase proximity outcome, not counterfactual task success.

The likely shortcut is structural in the dataset. Each scene template is paired with an almost fixed target, so scene identity can substitute for object-name grounding. Decorrelated scripted demonstrations improve the counterfactual metric, but all three fine-tuning attempts severely reduce ordinary closed-loop success. The best repaired dataset raises capability only to 21.5%, far below the healthy range.

This mixed result narrows the VLA claim. Language changes the trajectory reproducibly, but scene priors often prevent complete rerouting. The model is a capable language-conditioned specialist, not yet a robust instruction-grounded policy. LIBERO-CF independently documents the same broad failure class in existing VLAs [10]. Our prospective novelty is therefore not the shortcut alone. It is the harder test of whether exact weight-derived terms can localize and selectively repair that shortcut.

6 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The same-skeleton control has one completed seed, a 4.4 to 5.8 point capability gap, and a residual compiled-runtime tax. The benchmark is one specialist suite, counterfactual instructions do not fully reroute the policy, and exact decomposition is layerwise rather than end to end. The closed-loop subspace test uses one checkpoint and twenty trials, while the first direct coefficient edit fails its selectivity gate. The defensible result is therefore narrow: rational restrictions do not prevent competitive LIBERO-Object control, and they provide a precise weight-derived analysis surface. Broader grounding, multi-seed controls, optimized execution, and validated coefficient edits remain necessary.

Reproducibility. The final artifact will provide code, configurations, checkpoint hashes, exact certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are unavailable.

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A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

248 A.2 Matched architecture capability and systems cost

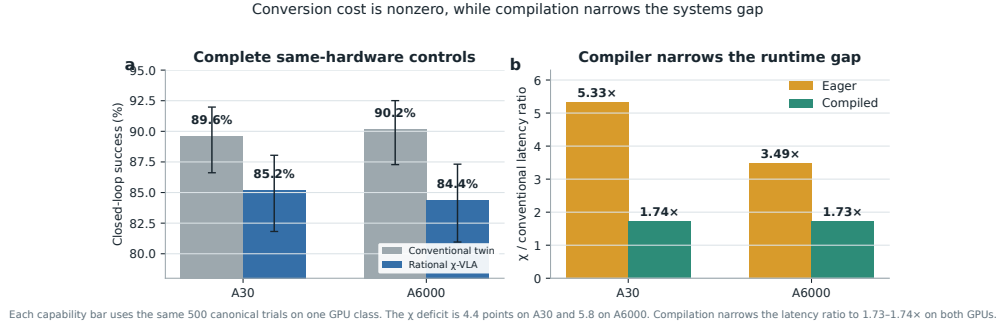


Figure 2: **Conversion cost is modest but nonzero, and compilation narrows the systems gap.** Complete same-hardware evaluations give 89.6% versus 85.2% on A30 and 90.2% versus 84.4% on A6000, conventional then rational. Parameters differ by 0.006%. Compilation narrows the batch-one latency ratio to 1.73–1.74x on both GPUs.

249 A.3 Three-checkpoint counterfactual grounding

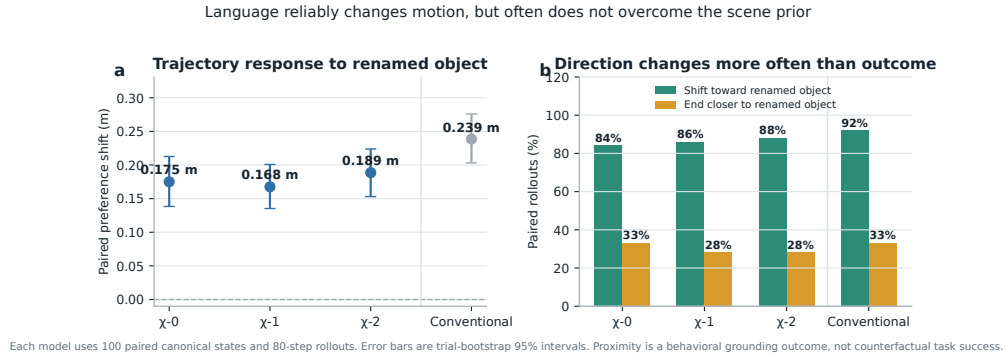


Figure 3: **Renaming the target changes motion reliably but does not fully override the scene prior.** Each model is evaluated on 100 paired canonical states for 80 control steps. The left panel reports the change in end-effector preference for the newly named object, with 95% bootstrap intervals. The right panel separates directional response from the stricter condition of ending closer to that object. These are proximity tests, not counterfactual task-success measurements. The conventional control shows that ordinary language response is not unique to tensor decomposability.

250 A.4 Causal visual-subspace intervention

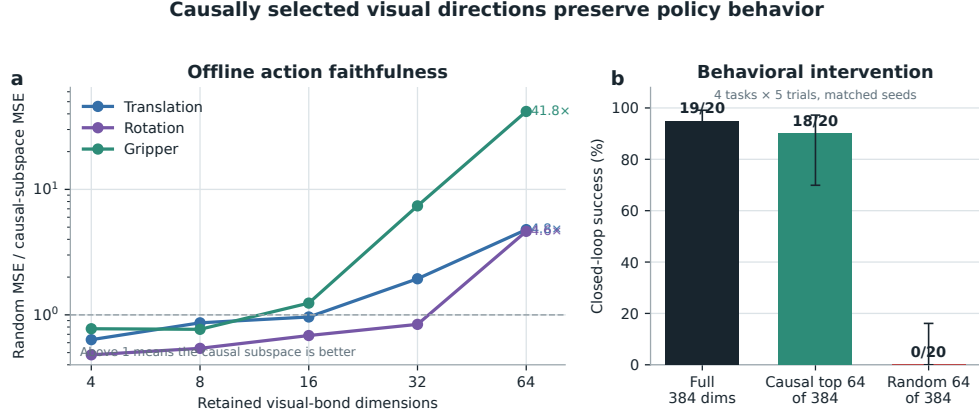


Figure 4: **Causally selected visual directions preserve policy behavior.** The left panel reports random MSE divided by selected-subspace MSE. The right panel intervenes on the real policy over four tasks and five paired trials per task.

251 A.5 Per-task ensemble behavior

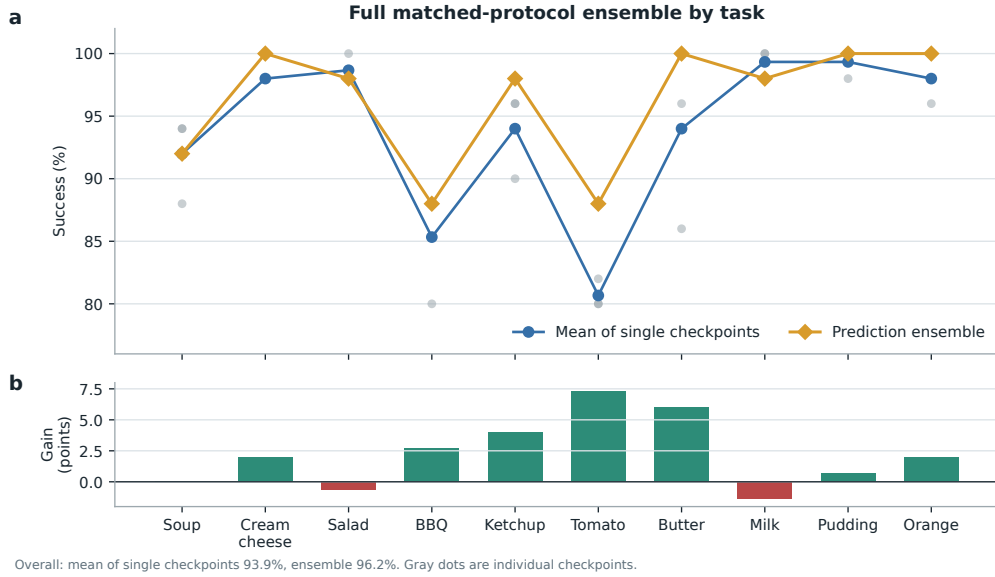


Figure 5: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

252 A.6 Product-routing multimodal head

253 For an action-query state h , the product-routing head represents

$$a(h, b) = c_0(h) + \sum_{g=1}^G (2b_g - 1)c_g(h), \quad b \in \{0, 1\}^G. \quad (4)$$

254 The center, offsets, and gates are bilinear CP modules. A final external sign choice selects one of
 255 2^G modes. The head folds to numerical precision, but its five-seed closed-loop mean is $44\% \pm 30\%$
 256 and its range overlaps the linear baseline. The strongest rational policy therefore uses the linear head.
 257 Product routing remains a construction result until it wins a controlled genuinely multimodal task.

258 A.7 Development interventions

259 Focused DAgger on BBQ and tomato improves combined success from 82.5% to 96.0% using seven
 260 corrective episodes and 1,131 frames. Applying the same update thinly across all ten tasks reduces
 261 overall success from 92.0% to 72.0%. Three original AWR runs decline by 8.0, 4.6, and 2.6 points.
 262 A reward audit shows that accumulated step cost makes every successful trajectory have negative
 263 return. After removing that term and repairing the terminal boundary, one pilot improves by two
 264 points. These are development diagnostics, not validated paper contributions.

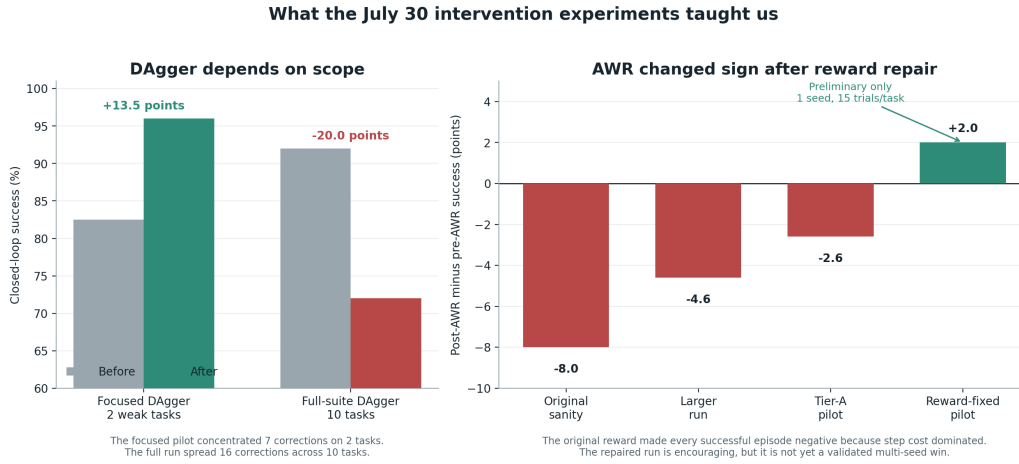


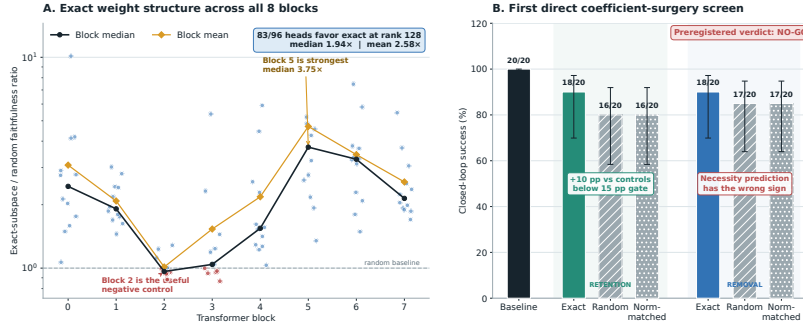
Figure 6: Development interventions reveal scope and reward sensitivity. Focused DAgger helps two weak tasks, while spreading a small correction budget across ten tasks causes regression. Three AWR variants also regress before a reward audit and repair. The positive reward-repaired AWR result is a one-seed pilot with 15 trials per task, so it is not evidence of a validated improvement.

265 A.8 Expanded attention screen and first direct edit

266 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 267 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 268 is 2.58, and blocks 2 and 3 provide useful weak or negative controls. At ranks 8 and 32, only 54/96
 269 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at
 270 the wider tested rank.

271 The first direct weight-edit screen uses four tasks and five paired canonical episodes per task. Exact-
 272 subspace retention preserves 18/20 successes compared with 16/20 for both random controls. This
 273 10-point advantage misses the preregistered 15-point gate. Exact removal also preserves 18/20,
 274 compared with 17/20 for both random controls, which is the wrong direction for the necessity
 275 prediction. The screen is a no-go for the current mapping from Gram eigenspaces to query, key, and
 276 value column edits.

Weights expose reproducible structure, but the first direct edit is not yet selective



A: 96 block-head pairs, fixed random-control banks, rank 128. Exact directions come from weights; cached activations only score layer faithfulness. B: 4 tasks $\times$ 5 paired canonical episodes. Direct Q/K/V weight edits, no hooks, gradients, retraining, or rollout fitting. Error bars are 95% Wilson intervals.

Figure 7: **Expanded weight-structure and direct-surgery results.** Left: all 96 block-head pairs at rank 128, with fixed random-control banks. Exact directions come from weights, while cached activations only score layer faithfulness. Right: the first direct query-key-value column-edit screen over four tasks and five paired trials per task. Error bars are 95% Wilson intervals. The preregistered verdict is no-go because retention misses the effect-size gate and removal has the wrong sign.

277 A.9 Numerical scope of rational conversion

278 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 279 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 280 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 281 establish practical numerical safety under distribution shift.

Mechanical decomposability audit of the trained 189/200 checkpoint

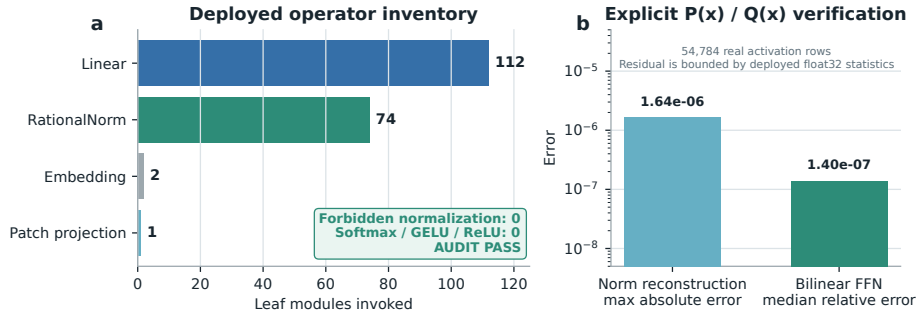


Figure 8: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.